

Troubleshooting

Symptoms, likely causes, and resolutions. The guiding principle: the system tells you the truth about its state — read the reported state/quality before assuming a bug.

Startup

“InsecureConfigError” / refuses to start - Cause: AIRONE_JWT_SECRET is unset, too short, or a placeholder. - Fix: export AIRONE_JWT_SECRET="`<>=32 random chars>`" and restart.

No telemetry appears (live mode) - Cause: no serial device, wrong port, or nothing transmitting. - Diagnosis: check the receiver’s reported port error in the logs. The system will **not** fabricate telemetry. - Fix: set telemetry.serial_port, verify wiring/baud, or run --simulate.

GUI does not open - Cause: no DISPLAY (headless) or PyQt5 not installed. - Behaviour: the launcher reports this and continues headless — this is expected, not a crash. - Fix: install PyQt5 and run on a machine with a display, or use the API.

Data quality

Values shown red/orange in the GUI - These are INVALID / STALE / SUSPECT measurements. The colour is the system telling you not to trust the number. Investigate the sensor, not the GUI.

A measurement is CORRUPTED - The frame failed CRC (and FEC could not recover it, or reedsolo is not installed). Check RF link quality; install reedsolo to enable FEC recovery.

An analysis says “insufficient data” - Expected behaviour when too few valid samples exist. Let more telemetry accumulate; the analysis will compute once it has enough.

ML tier

/ml/anomalies returns 503 NOT_CONFIGURED - The ML worker is not running. Ensure the orchestrator started it (it is wired automatically at startup). This is honest reporting, not an error to paper over.

A detector shows available: false - Its optional dependency is missing. isolation_forest/gmm need scikit-learn; lstm_forecast needs PyTorch. robust_zscore always works. - Fix: pip install scikit-learn and/or the CPU build of torch.

Training returns 422 - Either there is no stored telemetry yet, or fewer than 20 valid samples after dropping incomplete frames. The response states the exact counts. Accumulate more data and retry.

SafeLoadError when loading a model - The artefact's SHA-256 does not match (tampered/corrupted) or no expected hash was provided. Retrain to produce a fresh, verified artefact.

API

401 Unauthorized - Missing/expired token. Log in again or refresh.

403 Forbidden - Your role is too low for that endpoint. See [security_model.md](#).

429 Too Many Requests - Rate limit hit. Back off; adjust `api.rate_limits` if the limit is too tight for your deployment.

Tests

GUI smoke tests fail with a Qt platform error - Run with `QT_QPA_PLATFORM=offscreen`. On CI, also install `libgl1 libegl1 libxkbcommon0 libdbus-1-3`.

Security tests fail on import - Ensure `AIRONE_JWT_SECRET` is set in the environment before running `pytest` (the shared `conftest.py` sets a test default automatically).